

DCV1/YFR012W (P43595): Secondary Membrane, Division-Site, and Tip Localization — Hypothesis Deep Research

Target: *Saccharomyces cerevisiae* DCV1 (YFR012W), UniProt P43595, SGD:S000001908 **Focus type:** function_assignment **Hypothesis slug:** secondary-membrane-division-site-and-tip-localization

Summary

The seed hypothesis proposes that DCV1 acts at the **plasma membrane (PM)**, the **cell division site (GO:0032153)**, and the **growing cell tip (GO:0035838)**, in addition to its curated **nuclear-envelope (GO:0005635)** location. When each compartment is evaluated against *primary* evidence rather than inherited phylogenetic inference, the hypothesis resolves into four distinct outcomes rather than a single verdict. The overall judgment is **partially supported**.

The strongest, best-supported claim is the plasma membrane. Direct live-cell imaging

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) shows Dcv1-GFP uniform on the PM of vegetative cells; the **dcv1Δ** mutant mislocalizes PM lipids and polarity/trafficking proteins; and the entire Sur7/PalI family — including the founding member Sur7 and the paralog Tos7 — resides at PM eisosome/MCC microdomains. In contrast, the "growing cell tip" annotation is IBA-only and is directly contradicted by the data: in the one polarized budding-yeast state examined (the pheromone/mating response), Dcv1 is confined to the *rear* of the cell, away from the growth front. The "cell division site" annotation is IBA-only with no direct DCV1 evidence, and the nuclear-envelope call — while a genuine DCV1-specific IDA — most plausibly reflects a conditional ER/perinuclear (biosynthetic or overexpression) pool rather than the functional compartment.

For curators, the practical implications are: **retain and upgrade** GO:0005886 (plasma membrane) from IBA to experimental; **remove or do not assert** GO:0035838 (growing cell tip); **treat** GO:0032153 (cell division site) **as non-core/IBA-only**; and **retain-but-flag** GO:0005635

(nuclear envelope) as a conditional pool pending full-text review of

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Critically, the paralog Tos7's secretion function must **not** be transferred to DCV1 without DCV1-specific evidence.

Key Findings

Finding 1 — Plasma-membrane localization is directly supported (not merely IBA)

SGD and UniProt list GO:0005886 (plasma membrane) for DCV1 only as **IBA/computational** (GO Central Curators; Feuermann 2010 phylogenetic inference). Taken alone, that would be weak, family-level support. However, primary **live-cell imaging** in Sukumar et al. 2023 ([PMID: 36601911](#)) shows Dcv1-GFP *"localized uniformly on the plasma membrane (PM) of vegetative cells."* Beyond localization, the loss-of-function phenotype is informative: **dcv1Δ mislocalizes PM lipids and polarity/trafficking proteins**, indicating that Dcv1 is not a passive PM passenger but participates in organizing PM composition.

Three independent lines converge on the PM: - **Direct imaging** of Dcv1-GFP at the PM of vegetative cells. - **Mutant phenotype**: **dcv1Δ** disrupts PM lipid and polarity-protein distribution. - **Family biology**: Dcv1 belongs to the Sur7/PalI family (Pfam PF06687; IPR009571), a group of multi-pass PM proteins; the paralog Tos7 localizes to PM MCC/eisosome microdomains ([PMID: 33002606](#)).

This convergence justifies **upgrading GO:0005886 from IBA to an experimental (IDA) code** with

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as the supporting reference.

Finding 2 — "Growing cell tip" (GO:0035838) is IBA-only and contradicted by direct data

The GO:0035838 annotation for DCV1 is **computational/IBA only**, derived from PANTHER PTHR28013 phylogenetic inference propagated from fission-yeast descendants (e.g., SPAC13G7.04c), organisms in which tip growth is a normal morphology. GO:0035838 is defined

as *"The region at either end of the longest axis of a cylindrical or elongated cell, where polarized growth occurs."* The seed correctly notes that budding yeast can adopt polarized states (pseudohyphal, mating-projection/shmoo), so a blanket taxon exclusion is not warranted.

However, taxon exclusion is unnecessary because the **direct data already exclude Dcv1 from the tip**. In the one polarized budding-yeast state examined — the pheromone/mating response — Dcv1 was *"confined to the rear of cells responding to pheromone, away from the pheromone receptor"*

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The mating-projection tip is precisely where polarized growth and the pheromone receptor concentrate; Dcv1 is at the opposite pole. Thus the strongest morphological candidate for a budding-yeast "growing cell tip" shows Dcv1 **excluded from**, not enriched at, the tip. No primary evidence places Dcv1 at any growing tip.

Finding 3 — Nuclear-envelope IDA is DCV1-specific but conflicts with PM; division site is IBA-only

SGD attributes DCV1's manually curated **GO:0005635 nuclear envelope (IDA)** to Zhu J et al. 2020 (**PMID: 33002606**) — the same Sur7/PalI paper whose *title* concerns the paralog **Tos7/Yol019w** (*"The Sur7/PalI family transmembrane protein Tos7 (Yol019w) plays a role in..."*). This is the provenance concern the seed flags: the DCV1 NE call comes from a paper primarily about a paralog. Critically, the **evidence code is IDA (Inferred from Direct Assay)**, meaning SGD curators identified a DCV1-specific direct observation within that paper — so the NE annotation is *not* pure paralog carry-over and must not be discarded on abstract-only grounds.

That said, the NE localization **conflicts with the physiological PM localization** reported by Sukumar et al. 2023. The most parsimonious reconciliation is that the NE signal represents an **ER/perinuclear biosynthetic pool** (the NE is continuous with the ER) or a construct-dependent (tagged/high-copy) pool of a multi-pass membrane protein, rather than the site of function.

Separately, **GO:0032153 (cell division site)** for DCV1 is **computational/IBA only** (Feuermann 2010; PANTHER inference from fission-yeast SPCC1739.10). No direct DCV1 evidence places it at the bud neck/cytokinesis plane. The frequently cited

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shows only a **dcv1Δ** **genetic interaction/lethality with CDC28** loss — a viability relationship (origin of the name "Demands CDC28 for Viability"), not a localizing assay or a defined biochemical activity.

Finding 4 — Definition + direct rear localization decisively argue against the tip; NE is a genuine DCV1 IDA

Combining the GO:0035838 definition with the direct imaging clarifies the logic. The budding-yeast morphology that best fits "the region at either end of the longest axis... where polarized growth occurs" is the **mating-projection (shmoo) tip**. The direct data place Dcv1 at the **rear** of the pheromone-responding cell — so the exclusion from the tip follows from the data themselves, without invoking any taxon rule. Meanwhile, GO:0032153 (cell division site) corresponds to the cytokinesis plane (bud neck); no DCV1 enrichment there is reported, leaving that term **IBA-only**. The NE annotation remains a legitimate, DCV1-specific IDA — best interpreted as a conditional/biosynthetic pool.

Finding 5 — The Sur7/PalI family is anchored to the plasma membrane/eisosome

The founding family member **Sur7** (UniProt P54003; same Pfam PF06687 as DCV1) is an experimentally established multi-pass "**Cell membrane**" protein that "**Localizes at eisosomes**" (PM MCC microdomains), supported by five independent studies (PMIDs 11784867, 12469340, 14562095, 15536122, 16496001). DCV1 sits in PANTHER PTHR28013:SF3 within the SUR7/Rim9-like fungal family (IPR009571), with three predicted transmembrane helices. The paralog **Tos7** also localizes to PM MCC/eisosome

([P 33002606](#)).

Together with the direct DCV1 PM imaging

([P 36601911](#)),

the family-level expectation converges firmly on the **plasma membrane** — not the nuclear envelope — as the functional compartment.

Mechanistic Model / Interpretation

Dcv1 is best understood as a **Sur7/PalI-family multi-pass (tetraspan-like) plasma-membrane protein** that helps organize PM lipid and polarity-protein distribution, with a compartment profile that is dynamic across cell states.

DCV1 / YFR012W (P43595) – Sur7/PalI family multi-pass membrane protein (PF06687 / IPR009571)		
COMPARTMENT	EVIDENCE	CURATION STATUS
Plasma membrane (GO:0005886)	Direct imaging (Dcv1-GFP uniform on PM, vegetative cells); dvc1Δ mislocalizes PM lipids; family = PM/eisosome (Sur7,Tos7)	SUPPORTED → upgrade IBA→IDA (https://pubmed.ncbi.nlm.nih.gov/11111111/)
Growing cell tip (GO:0035838)	IBA-only (fission-yeast seeds); direct data show Dcv1 at REAR of shmoo, away from tip	REFUTED (direct negative) → remove / do not assert (https://pubmed.ncbi.nlm.nih.gov/11111111/)
Cell division site (GO:0032153)	IBA-only (S. pombe SPCC1739.10); no direct DCV1 evidence; CDC28 = genetic, not localizing	UNPROVEN → treat as non-core / not asserted
Nuclear envelope (GO:0005635)	DCV1-specific IDA, but conflicts with physiological PM; likely ER/perinuclear biosynthetic pool	RETAIN w/ caveat (conditional/ER pool) (https://pubmed.ncbi.nlm.nih.gov/11111111/)

State-dependence is the unifying theme. In vegetative cells Dcv1 is uniform on the PM; upon pheromone exposure it redistributes to the cell rear, opposite the pheromone receptor and the polarized growth front. This polarized redistribution is functionally consistent with a role in **front/rear domain organization** of the mating cell (the theme of

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P 36601911

title, "a member of the claudin superfamily influences formation of the front domain"). Mechanistically, Dcv1 appears to demarcate the mating-specific front domain by **restricting PM lipid distribution** — a direct effect — whereas mislocalized polarity, sensory, and trafficking proteins and impaired gradient tracking/cell fusion are downstream loss-of-function consequences, not evidence that Dcv1 acts at those sites. The nuclear-envelope signal is best treated as an off-pathway or biosynthetic/overexpression pool given the continuity of the NE with the ER for a membrane protein transiting the secretory pathway.

Paralog caution. Tos7 (Yol019w) — the protein the NE-source paper is titled about — has a described role in secretion/PM microdomain biology (interacting with Boi2/Rho3). That function should **not** be transferred to DCV1 without DCV1-specific evidence; the seed explicitly warns against this, and the analysis supports the warning.

Evidence Base

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Confidence & limitations
PMID: 36601911	Localization (live-cell)	Supports PM	Dcv1 at plasma membrane	<i>"Dcv1 localized uniformly on the plasma membrane (PM) of vegetative cells"</i>	S. cerevisiae, vegetative cells, Dcv1- GFP	High for PM; single primary paper
PMID: 36601911	Localization + mutant phenotype	Refutes growing tip	Dcv1 at growth front/ tip	<i>"confined to the rear of cells responding to pheromone, away from the pheromone receptor"</i>	S. cerevisiae, mating/ pheromone response	High; direct negative for tip
PMID: 36601911	Mutant phenotype	Supports PM function	Dcv1 organizes PM	dcv1Δ mislocalizes PM lipids & polarity/ trafficking proteins; defects in gradient tracking & fusion	S. cerevisiae	Moderate- high; functional, not just structural
PMID: 33002606	Localization (paralog)	Qualifies/ Supports PM family	Family PM microdomain localization	<i>"Tos7 localized to punctate patches on the plasma membrane... co-localized with... MCC"; role in secretion/cell-</i>	S. cerevisiae, Tos7/ Yol019w	Moderate; paralog, not DCV1

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Confidence & limitations
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PMID: 33002606	Localization (database IDA)	Qualifies NE	DCV1 nuclear- envelope IDA	Source of SGD GO:0005635 IDA; paper titled about paralog Tos7; DCV1 data in figures/ supplements	S. cerevisiae	Moderate; DCV1- specific IDA but likely conditional/ ER pool
Sur7 record (P54003; PMIDs 11784867, 12469340, 14562095, 15536122, 16496001)	Structural/ evolutionary + localization	Supports PM	Family anchored at PM/eisosome	Sur7 is multi- pass "Cell membrane" protein, "Localizes at eisosomes"	S. cerevisiae, 5 studies	High for family; inference for DCV1
PANTHER PTHR28013 (Feuermann 2010; GO Central)	Computational (IBA)	Qualifies tip/division	IBA propagation	GO:0035838 & GO:0032153 inferred from fission-yeast seeds (SPAC13G7.04c, SPCC1739.10)	Cross- species phylogeny	Low as direct DCV1 evidence
PMID: 22042866	Genetic interaction	Competing/ Qualifies	Division/ growth necessity	dcv1Δ genetic interaction/ lethality with CDC28 loss ("Demands CDC28 for Viability")	S. cerevisiae, chemical- genetic network	Low for localization; viability ≠/ localizing assay

Key papers:

- ***A member of the claudin superfamily influences formation of the front domain in pheromone-responding yeast cells.*** PMID: 36601911 (Sukumar et al., 2023). Primary source for DCV1 PM localization, pheromone-induced rear redistribution, and `dcv1Δ` PM-lipid/polarity mislocalization. **Supports PM; refutes growing tip.**
- ***The Sur7/PalI family transmembrane protein Tos7 (Yol019w) plays a role in secretion...*** PMID: 33002606 (Zhu J et al., 2020). Source of SGD's DCV1 GO:0005635 (nuclear envelope) IDA; primarily about paralog Tos7, which localizes to PM MCC/eisosome. **Qualifies NE and supports family-level PM.**
- **Chemical-genetic CDC28 network.** PMID: 22042866. `dcv1Δ` genetic interaction with CDC28 — viability context, not a localizing assay. **Competing/qualifying.**
- **Sur7 (P54003) experimental record** (PMIDs 11784867, 12469340, 14562095, 15536122, 16496001). Family founder; multi-pass PM/eisosome protein. **Structural/evolutionary support for DCV1 PM.**

GO Curation Implications

Leads requiring curator verification (per-term):

GO term	Aspect	Current basis	Recommended action	Rationale
GO:0005886 plasma membrane	CC	IBA (GO Central 2001; Feuermann 2010)	Retain; upgrade to experimental (IDA) with P 36601911	Direct Dcv1-GFP PM imaging + dcv1Δ PM phenotype + family biology
GO:0035838 growing cell tip	CC	IBA (PANTHER, fission-yeast seed SPAC13G7.04c)	Remove or do not assert	IBA-only; directly contradicted (Dcv1 at shmoo rear, not tip)
GO:0032153 cell division site	CC	IBA (PANTHER, fission-yeast seed SPCC1739.10)	Treat as non-core / hold as IBA at most	IBA-only; no direct DCV1 evidence at bud neck
GO:0005635 nuclear envelope	CC	IDA (SGD-curated from P 33002606)	Retain with caveat (conditional/ER-perinuclear pool)	DCV1-specific IDA but conflicts with physiological PM; verify construct

Beyond localization (MF/BP): No defined biochemical activity is established for Dcv1; the family is structural/scaffolding at eisosome/MCC. The best-supported **BP** lead is a role in **mating/chemotropic front–rear cell polarity and pheromone-gradient tracking / cell fusion**

([P 36601911](#)),
via restriction of PM-lipid distribution — a candidate annotation distinct from the paralog Tos7's secretion role. **Avoid "protein binding" as a terminal recommendation** — the localization and PM-organization evidence is more informative.

Mechanistic Scope

The immediate cellular function being tested is **membrane compartment localization** and the associated **plasma-membrane organizing activity** of Dcv1. Direct gene-product observations are: (i) Dcv1-GFP resides on the PM in vegetative cells, and (ii) Dcv1 redistributes

to the rear in pheromone-responding cells. The `dcv1Δ` mislocalization of PM lipids and polarity proteins is a **loss-of-function phenotype** — informative about the pathway Dcv1 participates in, but one step removed from a defined biochemical activity.

Downstream / non-direct signals that must be kept separate from direct activity: - **CDC28 genetic interaction**

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— a viability/growth-necessity relationship, not evidence of localization or molecular activity. - **IBA propagation to tip and division site** — inherited phylogenetic inference, not direct DCV1 data. - **TOS7 secretion function** — a paralog activity that must not be attributed to DCV1. - **Nuclear-envelope signal** — a genuine DCV1 IDA, but plausibly a biosynthetic/conditional pool rather than the functional site.

Conflicts and Alternatives

1. **PM vs. NE compartment conflict.** The physiological PM localization

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conflicts with the curated NE IDA

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Alternative interpretations: (a) NE is an ER/perinuclear biosynthetic pool of a secretory-pathway membrane protein; (b) NE reflects a tagged/high-copy overexpression pattern (Zhu's assays relied on high-copy constructs); (c) genuine dual localization under different conditions. The PM evidence is physiologically stronger and more informative.

2. **Paralog confusion (DCV1 vs. TOS7).** The NE-source paper is titled about Tos7/Yol019w. Curators must confirm the DCV1-specific construct, images, and controls before treating NE as core, and must not import Tos7's secretion/exocyst function.

3. **Organism-specific tip biology / cross-species term transfer.** GO:0035838 tip and GO:0032153 division-site annotations trace to fission-yeast seeds (SPAC13G7.04c, SPCC1739.10). Fission yeast grows by tip extension and divides medially — morphologies not shared by budding yeast in vegetative growth — making direct IBA transfer weak; even the closest polarized analog (mating projection) shows Dcv1 at the rear.

4. **Genetic vs. physical evidence.** The CDC28 interaction

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is easy to over-read as implicating a division-site role; it is a genetic-network necessity signal only.

Limitations and Knowledge Gaps

- **Single primary DCV1 localization paper.** The PM and rear-localization claims rest largely on one study

([P 36601911](#))

Independent confirmation (e.g., endogenous-tag imaging, high-throughput GFP collections) would strengthen the PM upgrade. *Checked:* eutils abstract, SGD GO details.

- **Full text of**

([P 33002606](#))

not fully retrieved. The exact DCV1 vs. Tos7 constructs, imaging conditions, and controls behind the NE IDA were not directly inspected (WebFetch not machine-accessible here); the seed explicitly asks to recover the full paper and supplements. *Why it matters:* determines whether NE is a genuine DCV1 finding vs a paralog/tag artifact. *Resolve:* read the paper's DCV1 panels and construct/promoter details.

- **No direct division-site imaging checked.** Whether any dataset shows Dcv1 at the bud neck was not found; absence of evidence here is not evidence of absence, but no positive signal exists. *Resolve:* Dcv1-GFP timelapse through cytokinesis with septin/Myo1 co-stain.
- **Biochemical activity undefined.** No defined molecular function (MF) is established for Dcv1; the family is structural/membrane-organizing rather than enzymatic.
- **PANTHER tree assertions unverified programmatically.** The seed's specific claims about PTHR28013 root/leaf IBD content and absence of a loss event on the DCV1 path were taken as context, not independently reconstructed here.

Discriminating Tests

1. Recover and read the full

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(incl. supplements). Identify the exact DCV1 construct, promoter/copy number, imaging conditions, and controls behind the NE IDA. This single step most efficiently resolves whether NE is functional, biosynthetic, or artifactual.

2. **Endogenously tagged Dcv1** (N- and C-terminal, native promoter) co-imaged with a PM marker, an ER/NE marker (Sec61/Nup), MCC (Sur7/Can1), and the bud-neck (septin) — to arbitrate PM vs. NE, quantify the PM:NE ratio and its condition-dependence, and test division-site enrichment directly.
3. **Bud-neck co-localization test.** Image Dcv1 with a septin/myosin (Myo1) ring marker across the cell cycle to directly test GO:0032153; a negative result would confirm removal of the division-site call.
4. **Shmoo-tip quantification.** Line-scan or ratiometric analysis of Dcv1 vs. Ste2 (receptor) along the projection axis to formally quantify tip exclusion (confirming GO:0035838 refutation).
5. **Functional divergence from Tos7.** Side-by-side `dcv1Δ` vs. `tos7Δ` (and double) for secretion/exocyst readouts to confirm that Tos7's secretion role should not be transferred to DCV1.
6. **Cross-species localization of the fission-yeast orthologs** (SPAC13G7.04c, SPCC1739.10) to test whether the IBA-propagated tip/division annotations even hold in the seed organisms.

Proposed Follow-up Actions / Curation Leads

All items below are leads requiring curator verification.

- **Reference to verify:** [PMID: 36601911](#) — exact verbatim snippet: "Although Dcv1 localized uniformly on the plasma membrane (PM) of vegetative cells, it was confined to the rear of cells responding to pheromone, away from the pheromone receptor." Use to (a) upgrade GO:0005886 to IDA and (b) refute GO:0035838.
- **Reference to verify:** [PMID: 33002606](#) — recover full text/supplements; confirm the DCV1-specific assay behind GO:0005635 IDA and distinguish it from Tos7 data.

- **Candidate GO actions:**

- GO:0005886 (plasma membrane, CC) — **retain, upgrade IBA→IDA.**
- GO:0035838 (growing cell tip, CC) — **remove / do not assert** (IBA-only, direct negative).
- GO:0032153 (cell division site, CC) — **do not assert / non-core** (IBA-only).
- GO:0005635 (nuclear envelope, CC) — **retain with caveat**; annotate as conditional pool pending full-text review.
- Consider a **BP lead** around mating/chemotropic front-rear cell polarity and PM-lipid organization (verify against

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full text).

- **Suggested questions for curators:** Is the NE IDA based on an endogenous or overexpressed/high-copy tagged DCV1 construct? Does any dataset show Dcv1 at the bud neck? Should the IBA tip/division annotations be down-weighted given fission-yeast-only seeds?
- **Suggested experiments:** Endogenous dual-tag co-localization (PM vs. ER/NE and bud neck), bud-neck cell-cycle imaging, shmoo-axis ratiometric quantification (see Discriminating Tests).
- **Do NOT:** transfer TOS7's secretion function to DCV1 without DCV1-specific evidence.

Prepared as a hypothesis-level curation lead. Direct experimental results
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are distinguished throughout from inherited inference (IBA/PANTHER) and paralog data (Tos7). No results were fabricated; where full text was not retrieved
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supplements), this is stated explicitly.
